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Research Article

Vet Assistant: An Online Integrated Clinic Management System

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Abstract: Many people keep animals as pets for company. Pets are fun to spend time with, but taking care of them is also hard work. Veterinarians ensure a safe food supply for people by monitoring and maintaining the health of food-producing animals. The Vet Assistant: An Online Integrated Clinic Management System was developed to aid the business for the automation of process and transactions specifically to Dr. S & J Veterinary Clinic and Grooming Center to test the efficiency of the software through several test cases and observation. The system is capable of data storage and is responsible for quick access and retrieval of data and information. Prior to the development of the system, it is designed with an add, edit, delete, update and search functions that allows the user to input different types of entries such as information about clients, pets, products, services & diagnosis findings for every checkup, that can also be edited and searched by keywords. Also, it is capable of handling point of sales with the integration of data from inventory stock in and stock out. For scheduled patient, the system can also notify clients via SMS without manually texting them one by one. As it was consulted by the end user or the veterinarian, the client's profile must contain all the information of its pet and its medical records stored in the system. The results obtained from testing cases shows that the proposed system offers and effective and efficient means in doing research activities for the researcher to deliver such system. And that the system also has passed the McCall's criteria which means that the system is of average quality.

Keywords: Clinic Management System, Integrated, Veterinary.

INTRODUCTION

Veterinary medicine is the branch of medicine that deals with the prevention, diagnosis and treatment of disease, disorder and injury in animals. A Veterinarian protects the health and well-being of both animals and people. Most pets with breeds are high risk in diseases if not being checked. Each owner owns a pet, and sometimes they own more than just one pet. For record keeping, the owner's pet data is being kept by the clinic. The clinic manages transaction of diagnostic information, products and services. To grow more profit, it is necessary for the clinic to monitor their sales.

According to Dr. Rocel S. Cantila-Sanogal (November 10, 2018) one of the owners and a veterinarian, they are having the difficulty of record archiving to their client's pet, they search it manually by the use of their printed forms which consist of an ascending pet number. In scheduling for the next appointment or next check-up of the pet, it is written in the pet's record form and manually notified by the front desk personnel via text message. The monitoring of their sales and inventories are done manually, they used a piece of paper to list down their items inside the clinic, that includes medicines, dog food, grooming items. In their point of sales, they do have their own receipts that has two copies, one for the client and one for their own record. They keep it so that by the end of the month or week they will sum up their sales for their income report.

The VET ASSISTANT: AN ONLINE INTEGRATED CLINIC MANAGEMENT SYSTEM (VAAOICM) is designed for the veterinary clinic located at Brgy. Alijis, Bacolod City, Negros Occidental. Intended to provide the integration in managing the clinic for the convenience and speed of the flow of transaction of the business.

The system plays a significant role in assessing the clinics management of their business. The feature of the system is to provide a database for data storing and an easy access of searching records. Also, it is capable of printing receipts and reports for the business, it also manages products inventory and supplier's delivery. The system also provides daily appointment scheduling reminders for check-up and an automatic notification to the client via use of SMS. It also has an informational home page which can be access and be viewed online. All for the purpose of providing a quality system for the satisfaction of our client.

OBJECTIVES OF THE STUDY

This study aimed to develop a system intended to the veterinary clinic for the automation of process inside the business entitled "VET ASSISTANT: AN ONLINE INTERGRATED CLINIC MANAGEMENT SYSTEM" that will aid all the transactions involved.

It specifically aimed to determine the quality of the developed system based on the McCall's Software Quality Model Criteria, evaluate the system based on Computer System Usability Questionnaire, determine the level of performance of the new system, develop a system that will automate the recording of data in information, sales and inventory management with generation of reports.

MATERIALS AND METHODS

Rapid Application Development Model was used in developing the "VET ASSISTANT: AN ONLINE INTERGRATED CLINIC MANAGEMENT SYSTEM", for the reason that it is a development model that prioritizes rapid prototyping and quick feedback over long-drawn-out development and testing cycles. With RAD, the researcher can make multiple iterations and updates to my software rapidly without needing to start a development schedule from scratch each time.

This research uses descriptive method to assess the gaps of the business and identify the needs and features of the system that will help the automation process of the clinic.

To have an idea of the old system, the researcher conducted the actual observation and followed by staff and doctors' interview regarding the flow and problems in the clinic. The respondents provided the relevant information needed by the researcher to create the idea and the functionalities and features that will be the key in the development of the system. After all of the data gathering, use case was created for every user that are involve in this research. Also, a work breakdown structure was designed for each area of the development of the system.

To identify the number of users, the researcher used Convenience Sampling Method and Purposive Sampling Technique.

Table 1.0 Respondents of the Study

Respondents	Number of users
Doctor	2
Front desk/Staff	1
Total respondents	3

Table 1.0 shows the sample size used for this study which is three (3) persons composed of the following: doctor (2) and front desk/staff (1). The purposive sampling technique was used to determine the sample size for all users.

In business observation and initial client interview with Dr. S & J Veterinary Clinic and Grooming Center, self-made questionnaire was used by the researcher.

Part I determined the quality of the developed system based on McCall's Software Quality Model.

Part II evaluated the system based on Computer System Usability Questionnaire.

Part III determined the level of performance of the new system.

Part IV reports generation assessment.

In determining the validity of the instrument, the researcher used Good and Skates, rated by fourteen random persons (14) including the three IT Expert, and one (1) Grammarian. The result was four-point twelve (4.12) which was interpreted as a valid questionnaire.

With the developed system, it was beneficial to the Dr. S & J Veterinary Clinic and Grooming Center to both admin or doctor and staff, because it aids to the automation of the business process including information management, sales and inventory management. It enables the business to have a more efficient, reliable and fast transactions of the clinic.

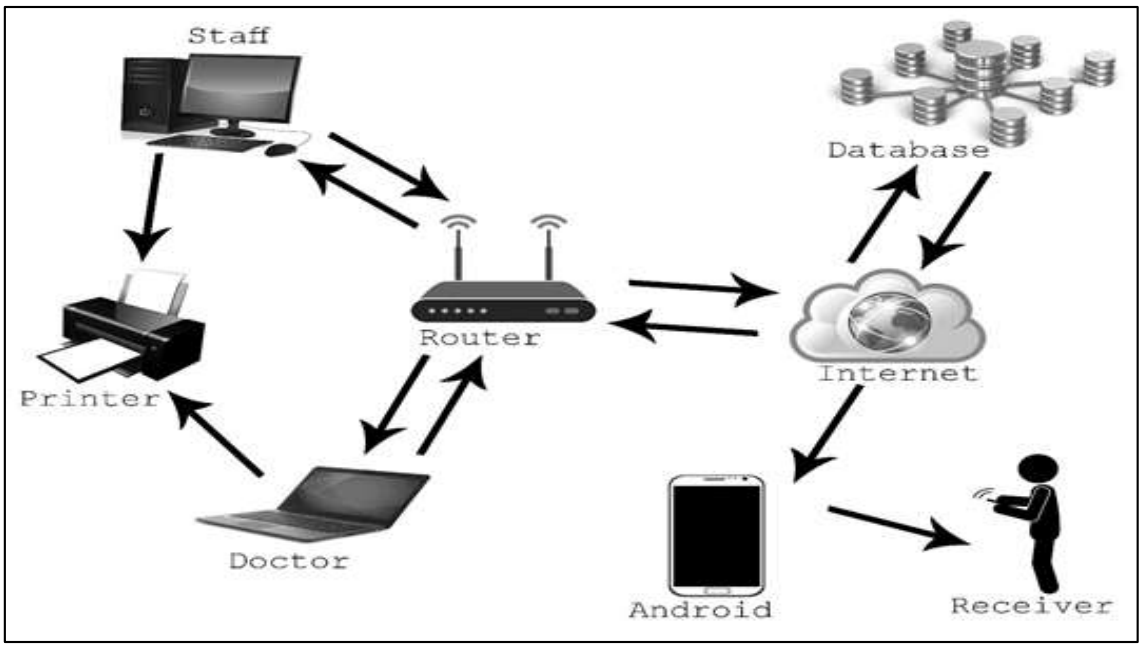

Figure 1.0 Architectural Design

Figure 1.0 show the design flow of the system architecture of the system where the concepts backbone of the actual system is developed. It is the conceptual model that describes the structure and behaviour of the proposed system or of an existing system. The model includes the technical framework, end user requirements, and a list of system components (hardware and software).

RESULTS AND DISCUSSION

To determine the level of quality of the developed system, three (3) IT Experts rated the system based on McCall's Criteria for Software Quality Model.

Table 2.0 McCall's Criteria for Software Quality Model

Criteria	Mean	Interpretation
Auditability	4	Good
Accuracy	4	Good
Completeness	5	Very Good
Communication Commonality	4	Good
Conciseness	4	Good
Consistency	5	Very Good
Operability	4	Good
Security	4	Good
Self-Documentation	4	Good
Software System Independence	5	Very Good
Traceability	4	Good
Training	4	Good
Data Commonality	5	Very Good
Error Tolerance	4	Good
Execution Efficiency	4	Good
Expandability	5	Very Good
Generality	4	Good
Hardware Independence	4	Good
Instrumentation	5	Very Good
Modularity	4	Good
Grand Mean	4	Good

Table 2.0 shows the results that the system has a Grand Mean of four-point ten (4) which is interpreted as Good based on the criteria set forth by McCall's Criteria for Software Quality. The result shows that the system was of good quality.

To determine the number of users, the researcher used the Purposive Sampling Technique for the admin or the doctor and the staff.

To determine the level of quality in the use of the developed system, the researcher used Computer System Usability Questionnaire (CSUQ) and was rated by the end-user.

Table 3.0 Level of Quality in Use of the Developed System based on CSUQ

	Mean	Interpretation
1. Overall, I am satisfied with how easy it is to use this system.	6.50	Agree
2.It was simple to use this system.	6.60	Strongly Agree
3. I can effectively complete my work using this system.	6.70	Strongly Agree
4. I am able to complete my work quickly using this system.	6.20	Agree
5. I am able to efficiently complete my work using this system.	6.50	Agree
6. I feel comfortable using this system.	6.30	Agree
7. It was easy to learn to use this system.	6.40	Agree
8. I believe I became productive quickly using this system.	6.70	Strongly Agree
9. The system gives error messages that clearly tell me how to fix problems.	6.60	Agree
10. Whenever I make a mistake using the system, I recover easily and quickly.	6.40	Agree
11. The information (such as online help, on-screen messages, and other documentation) provided with this system is clear.	6.40	Agree
12. It is easy to find the information I needed.	6.70	Strongly Agree
13. The information provided for the system is easy to understand.	6.80	Strongly Agree
14. The information is effective in helping me complete the tasks and scenarios.	6.50	Agree
15. The organization of information on the system screens is clear.	6.60	Strongly Agree
16. The interface of this system is pleasant.	6.50	Agree
17. I like using the interface of this system.	6.60	Strongly Agree
18. This system has all the functions and capabilities I expect it to have.	6.30	Agree
19. Overall, I am satisfied with this system.	6.40	Agree
Grand Mean	6.51	Strongly Agree

Table 3.0 shows the result in determining the level of quality of the developed system, the system has a Grand Mean of Six Point Fifty-One (6.51) which is interpreted as Strongly Agree.

To determine the level of performance of the developed system, the researcher used Researcher's Self-made Questionnaire.

Table 4.0 Level of the Performance of the New System based on Researcher's' Self-made Questionnaire.

In Terms of Determining the Level of Performance of the New System.	**Mean**	**Interpretation**
For Admin/Doctor		
1. The system provides fast searching of client's profile	5.00	Very Good
2. The system provides a reliable section to perform diagnosis and check-ups.	5.00	Very Good
3. The system provides follow up check-up scheduling and one-day advance reminder to clients via SMS	5.00	Very Good
4. The system provides reliable management of services	5.00	Very Good
5. The system provides new users for new staffs	5.00	Very Good
6. The system provides reminders of payment for unpaid products.	5.00	Very Good
For Staff		
7. The system provides reliable storage for client's and pet's profile management.	5.00	Very Good
8. The system provides quick record searching for scheduled clients as well as list of all clients for reschedule.	5.00	Very Good
9. In terms of inventories, management of products is more convenient.	4.00	Good
10. The system efficiently generates receipts for services and products sold.	5.00	Very Good
11. The system assesses the deliveries of products and payment for each supplier.	4.00	Good
Grand Mean	4.81	Very Good
In terms of generating reports in sales, inventory and client's record.	Mean	Interpretation
For Admin/ Doctor		
1. The system is efficient in generating reports	5.00	Very Good
2. The reports generated by the system is reliable.	5.00	Very Good
3. The patient records generated by the system are exact.	5.00	Very Good
4. Threshold printing of reports on products that are low in stocks are accurate.	5.00	Very Good
Grand Mean	5.00	Very Good

Table 4.0 shows the result in terms of determining the level of the Performance of the Developed System, the system has a Grand Mean of four Point eighty-one (4.81) which is interpreted as Very Good. Based on the result, the system has provided the aid of managing the business in an automated approach, it helps the doctor and the staff to monitor every client, as well as the management of their sales and product inventory. In terms of generating reports such as receipts, client's profile, sales and inventory reports, the system has a Grand Mean of Five (5.00) and was interpreted as Very Good. Based on the result, the system provides correct and useful generation of reports needed for the clinic's business.

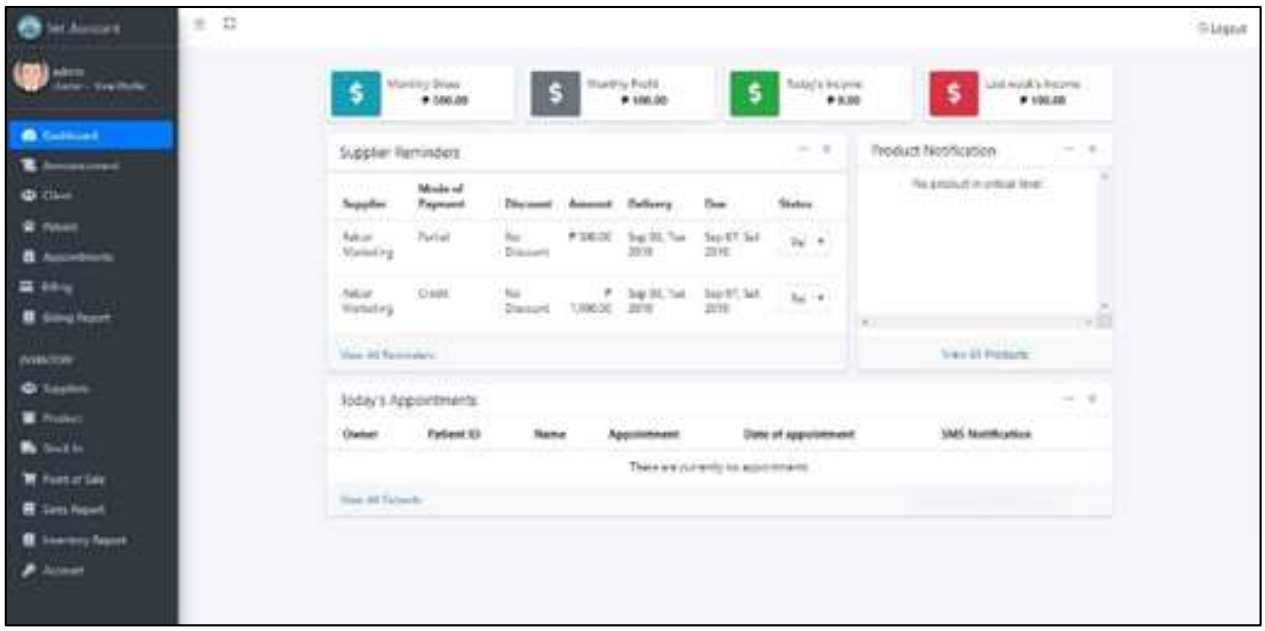

Figure 2.0 System Screenshot

Figure 2.0 shows the sample system screenshot of the system. The Dashboard of the system which contains the supplier reminders, appointments, notifications and other functions of the system in the side bar.

Conclusion

Based on the findings presented in this study, the following are the conclusions:

1. The system has passed the evaluation based on McCall's Criteria, thus considered as of good quality.
2. The system has passed all the characteristics from Computer System Usability Questionnaire (CSUQ); thus, it is considered as usable.
3. The system has upgraded the functionalities and capabilities of the business flow inside the clinic by using the developed system.
4. The system has upgraded the level of performance of the system currently used by the staff and doctor using the Researchers' Self-made Questionnaire.
5. The system has the capabilities of efficiently generating needed reports and forms by the clinic.

Recommendation

On the bases of the foregoing findings and conclusions, the following recommendations are submitted:

1. It is recommended to cope with the current flow of the clinic and the advancement of technology and growth of the business, the clinic may consider implementing the developed system.
2. It is also recommended that similar study must be conducted to assess further effectiveness and usefulness of the develop system.
3. In terms of the level of performance of the new system, the system was efficient thus recommended for utilization.
4. In terms of generating reports, the systems were effective thus recommended for implementation.

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